Bank Management System Using Java

A Core Java mini project: a menu-driven console application for basic banking





Introduction



Simple banking software

Handles basic banking work on a computer.



Saves time

Manual records and balance sums take longer.



Core Java, console based

An easy menu that beginners can follow.

Problem Statement

Develop a simple Java application for basic banking: account details, deposit, withdrawal and balance check.

Abstract

A menu-driven project that shows how Java solves a real-world task.

Objectives & Requirements

- Build a simple banking application
- Display customer account details
- Perform deposit operations
- Perform withdrawal operations
- Check the available balance
- Learn basic Java concepts

Hardware

- Computer or laptop
- Minimum 2 GB RAM
- Keyboard and monitor

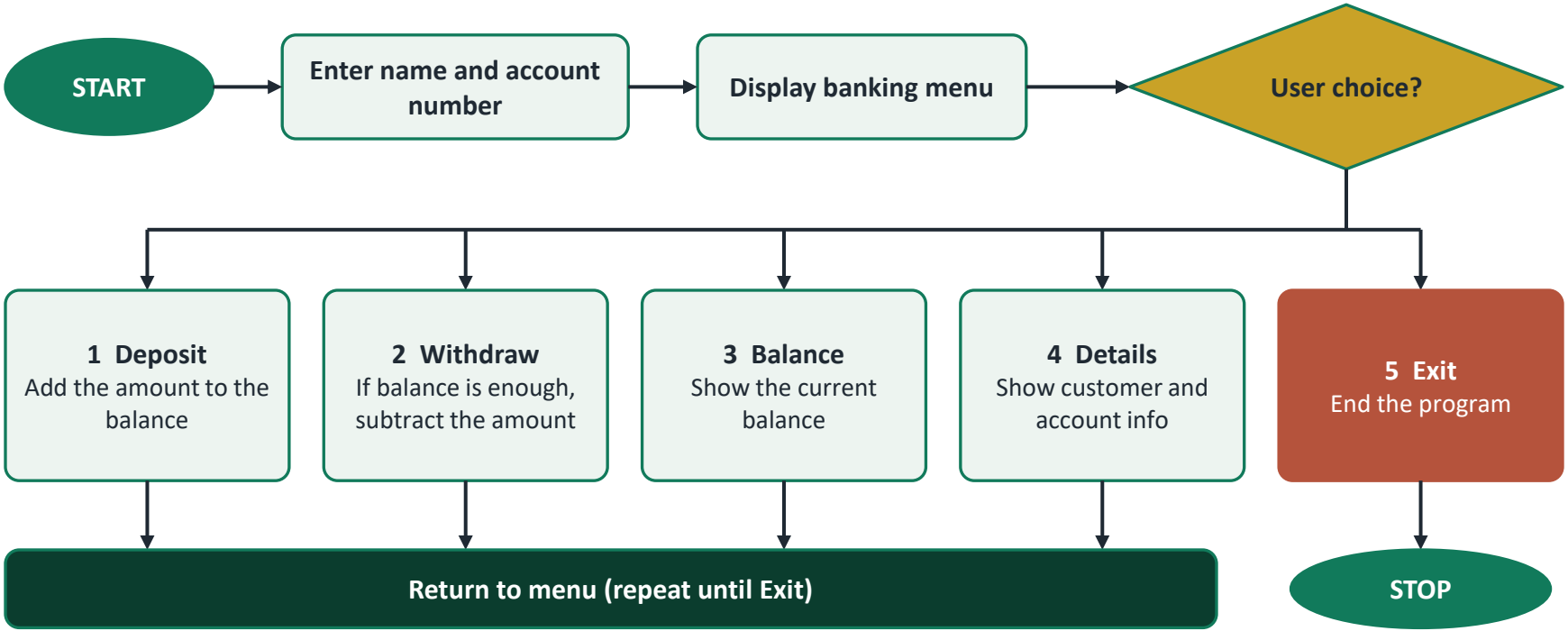
Software

- Java Development Kit (JDK)
- VS Code / Eclipse / IntelliJ
- Windows or Linux

Algorithm

1. Start and import the Scanner
2. Read customer name and account number
3. Show the banking menu
4. User selects an option
5. Deposit: add the amount to balance
6. Withdraw: if balance is enough, subtract amount
7. Balance / Details: display the information
8. Repeat menu until Exit, then stop

Flowchart


START
Enter name and account number
Display banking menu
User choice?
1 Deposit
Add the amount to the balance
2 Withdraw
If balance is enough, subtract the amount
3 Balance
Show the current balance
4 Details
Show customer and account info
5 Exit
End the program
Return to menu (repeat until Exit)
STOP

Methodology

Input module
Name, account number, choice, amounts

Processing module
Deposit adds, withdraw subtracts

Decision module
if-else and switch-case pick the task

Looping module
do-while repeats the menu until Exit

Output module
Details, messages and balance on screen

Java concepts used

- Scanner class
- Variables
- switch-case
- if-else
- do-while loop

Result: Sample Output

```
--- CUSTOMER DASHBOARD ---
1. Create Account
2. Deposit
3. Withdraw
4. Check Balance
5. View Transactions
6. Logout
Enter choice: 3
Enter account number: 1001
Enter withdrawal amount: 200
Withdrawal successful!
Remaining Balance: Rs. 300
```



Register and login
Create an account, then log in.



Withdraw money
Rs. 200 taken out. Balance left: Rs. 300.



Error message
A wrong password shows an error.

Conclusion & Future Scope

Conclusion

- All banking operations work: deposit, withdraw, balance and details.
- Practised Scanner, switch-case, if-else and do-while loops.
- Simple and friendly for Java beginners.

Future Enhancements

- Customer registration
- ATM / debit card
- Discount offers
- MySQL database
- Login system
- Swing / JavaFX GUI